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Procedia Computer Science 00 (2026) 000–000

Procedia Computer Science

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International Conference on Machine Learning and Data Engineering

A Dual-Transformer Fusion Framework for Robust Urban Environmental Sound Classification

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Abstract

Urban sound classifiers have to deal with two kinds of structure at once: the spectral shape of a sound and the way it unfolds in time. We pair an Audio Spectrogram Transformer (AST), which reads log-Mel patches, with a wav2vec 2.0 encoder that works directly on the waveform. Each branch turns the same clip into a 768-dimensional embedding, and a LightGBM classifier is trained on their concatenation.

On the ten official UrbanSound8K folds the system reaches 95.42% accuracy (macro-F1 0.955), with a five-run 95% confidence interval of 95.26 to 95.58%. It scores above AST, wav2vec 2.0, PaSST, HTS-AT and BEATs retrained under identical folds, though only narrowly above BEATs, and reaches 96.7% on ESC-50, where BEATs stays slightly ahead. The full pipeline needs a median of 184 ms per 4 s clip on an RTX 3060. To test the model beyond clean audio we built a Digital Twin of the acoustic path that combines distance-dependent propagation, reverberation, recorded urban ambience, UAV rotor noise and sensor effects. Accuracy falls from 95.42% on clean audio to 82.8% in the hardest scenario (50 m, −5 dB), and one-factor sweeps show that the signal-to-noise ratio drives most of that loss.

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering.

Keywords: Environmental sound classification; Audio Spectrogram Transformer; wav2vec 2.0; Embedding-level fusion; LightGBM; Digital twin; UAV acoustics

1. Introduction

1.1. Background and motivation

Smart-city platforms and autonomous sensing systems have created a pressing need for dependable, real-time awareness of the acoustic environment [1, 2]. City soundscapes are noisy in more ways than

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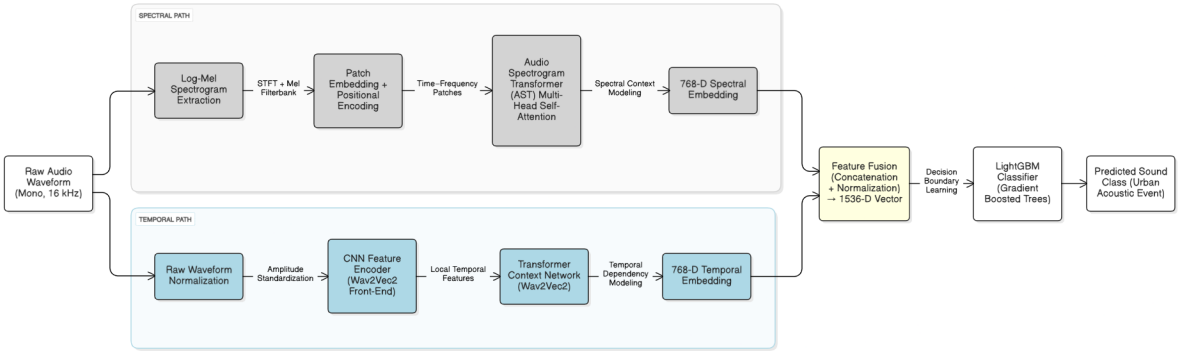


Fig. 1. Bird's-eye view of the dual-transformer fusion architecture. The spectral and temporal branches run in parallel on the same clip; their embeddings are standardised, concatenated and passed to LightGBM.

one: traffic, construction machinery and background chatter overlap with safety-critical cues such as sirens, alarms, gunshots and emergency-vehicle signals. These sounds often carry actionable information, especially when cameras are blocked by occlusion, smoke, poor light or dense crowds. Figure 1 gives a bird's-eye view of the architecture we propose. Raw audio is processed along two paths at once: a spectral branch built around an Audio Spectrogram Transformer and a waveform branch built on wav2vec 2.0. Both branches look at the same signal, so we treat them as complementary representations of one recording rather than as separate sensing modalities.

Older ESC pipelines extracted hand-designed descriptors, such as Mel-frequency cepstral coefficients (MFCCs), spectral contrast and zero-crossing rate, and passed them to shallow classifiers [3, 4]. These setups are fast but brittle, because they cannot represent the non-stationary, overlapping character of real urban sound events well. Deep learning changed the picture. CNNs operating on spectrogram images pushed accuracy up by a wide margin [5, 6], yet their local receptive fields still make long-range temporal patterns hard to capture.

Transformer models brought self-attention to audio and opened the door to global context across long time spans. AST and wav2vec 2.0 have each shown strong results, one on spectral structure and the other on waveform-level dynamics [7, 8]. The catch is that each sees the signal from only one side. That gap is what motivated us to build a two-branch framework that draws on both views at once.

1.2. Problem statement

Even with recent progress, several hard problems remain when ESC moves to real streets. Urban soundscapes shift constantly, and sources blend in time and frequency [9]. Models with limited temporal context struggle with events whose meaning lies in how they evolve: a burst of gunfire, a rising siren, a repeating jackhammer cycle [10]. Spectral transformers handle harmonic structure well, but their patch grid can blur very short onsets. Waveform encoders keep the timing, yet those pretrained on speech were not designed for broadband urban noise, and applying speech models directly to general audio is known to be suboptimal [11]. Many current models are also too heavy for real-time use on UAVs or edge devices. Our own system is not light either, so we measure its cost directly instead of assuming it.

A second problem is evaluation. Clean benchmark accuracy says little about how a classifier behaves when the source is far away, the street is loud and the microphone sits on a drone. Field campaigns that cover every combination of distance, noise and platform are slow and hard to repeat. We therefore use a Digital Twin, a parameterised model of the acoustic path in which each factor can be changed on its own while everything else stays fixed.

1.3. Contributions

This paper makes three contributions. First, we describe a simple and fully specified fusion of AST and wav2vec 2.0 embeddings with a LightGBM classifier, including the fine-tuning schedule, variable-length handling and leakage controls needed to reproduce it. Second, we compare it under identical folds with modern pretrained audio transformers (PaSST, HTS-AT and BEATs) on UrbanSound8K and ESC-50, report five-run confidence intervals and a paired significance test, and measure end-to-end latency, memory and model size. Third, we use a Digital Twin with recorded urban ambience to trace how accuracy degrades as distance and signal-to-noise ratio (SNR) worsen, and we separate the two factors with controlled sweeps.

2. Literature review

Research on environmental sound classification has moved through several distinct stages over the last decade, each shaped by the tools available at the time.

2.1. Handcrafted feature-based approaches

The earliest ESC work depended on hand-designed audio features: MFCCs, spectral roll-off, spectral centroid, bandwidth, zero-crossing rate and chroma vectors. These were paired with classifiers such as Gaussian mixture models, support vector machines, k-NN and random forests [3, 9]. The approach was light on computation but fell apart quickly in noisy or reverberant city conditions, where the features could not encode enough about complex, time-varying sounds.

2.2. Convolutional neural network-based ESC

Moving CNNs onto Mel-spectrogram images was a turning point [5]. Data augmentation, deeper VGG- and ResNet-style networks and large-scale pretraining each added measurable gains [6, 12, 13]. The weakness of these models is shared: convolutional kernels are local, so the effective field of view is narrow, and periodic or impulsive sounds that carry information over longer stretches are poorly represented.

2.3. Sequence models and recurrent architectures

Recurrent networks such as LSTMs and GRUs improved the temporal modelling of audio streams [14]. Hybrid CNN–RNN pipelines combined local spectrogram features with recurrent sequence processing [10], but training was slower, parallelisation limited and scaling difficult.

2.4. Transformer-based audio representation learning

The Audio Spectrogram Transformer follows the Vision Transformer [15]: it slices a log-Mel spectrogram into fixed-size patches and lets multi-head self-attention learn global spectral relationships [7]. With ImageNet and AudioSet [16] pretraining it outperformed earlier CNNs on ESC benchmarks. wav2vec 2.0 takes a different route: a convolutional front end feeds a transformer encoder that learns waveform representations through self-supervision [8]. Each of these single-stream designs has blind spots that the other could fill.

Since AST, stronger pretrained audio transformers have appeared. PaSST speeds up training by dropping patches [17], HTS-AT uses a hierarchical token-semantic design [18], and BEATs learns from discrete acoustic tokens and reports 98.1% on ESC-50 [11]. These published figures come from different pretraining data, input lengths and fine-tuning recipes, so they are not directly comparable with each other or with ours. For that reason we retrain these models under our own folds (Section 4.4) and cite published numbers only for context.

2.5. Fusion strategies for environmental sound classification

Fusion can happen at the input level (early), at the embedding level (intermediate) or at the decision level (late) [19]. Intermediate fusion is a common choice because it keeps the information in each representation while still letting a single classifier see both. Our framework uses plain embedding-level concatenation, which is cheap and easy to reproduce.

2.6. Simulation-based evaluation and digital twins

A digital twin is a virtual counterpart of a physical system that can be used to test behaviour before, or alongside, deployment [20]. In audio, controlled synthesis of soundscapes from isolated events and recorded backgrounds is an established way to build test conditions that would be impractical to record [21], and physically based simulation is increasingly used to study urban sound propagation itself [22]. We follow this line: our twin mixes real ambient recordings with simulated propagation, rotor and sensor effects so that each factor can be varied independently.

3. Proposed methodology

3.1. Overview of the dual-transformer framework

The design is a dual-stream pipeline. Spectral and temporal features are drawn from the same urban audio clip at the same time. The signal is resampled to 16 kHz and sent down two paths: a spectral branch that works on log-Mel spectrograms and a temporal branch that works on the raw waveform. Each branch outputs a 768-dimensional embedding; the two are standardised, concatenated and passed to a gradient-boosting classifier.

3.2. Datasets and evaluation protocol

UrbanSound8K. The corpus contains 8,732 clips of up to 4 s in ten classes: air conditioner, car horn, children playing, dog bark, drilling, engine idling, gunshot, jackhammer, siren and street music [9]. It ships with ten predefined folds, and clips cut from the same Freesound recording always share a fold. The dataset authors recommend these folds because reshuffling clips places near-identical excerpts on both sides of the split and inflates accuracy. We use them unchanged. Each fold k serves once as the test fold. Of the remaining nine, fold $(k + 1) \bmod 10$ is used for model selection and the other eight for training. Once hyperparameters, the number of epochs and the number of trees have been chosen without touching the test fold, both encoders and the classifier are refitted on all nine non-test folds and the test fold is predicted once. Every baseline follows exactly the same rule. Five seeds (11, 23, 37, 51 and 71) change initialisation and augmentation order but never the fold assignment.

ESC-50. As a second dataset we use ESC-50, which has 2,000 five-second recordings in 50 classes and five official folds [23]. One fold is used for testing, one of the remaining four for selection and three for preliminary training; the selected setting is then refitted on all four non-test folds. A separate 50-class head and a separate LightGBM model are trained on ESC-50 data; nothing learned for the ten UrbanSound8K classes is reused.

3.3. Leakage controls and reproducibility

We keep a manifest with the file path, Freesound source ID, fold, checksum and label of every clip. For each run we check that no path, source ID or file hash appears in both the training or validation partitions and the test fold, and we screen for near-duplicates. The feature scaler is fitted on the current training portion only, first during selection and again on all nine non-test folds for the final fit. Augmentation noise comes from a training-only library, and the noise recordings used in the Digital Twin test mixtures (Section 5) are disjoint from it. No epoch, checkpoint, tree count, threshold or scenario setting is chosen by

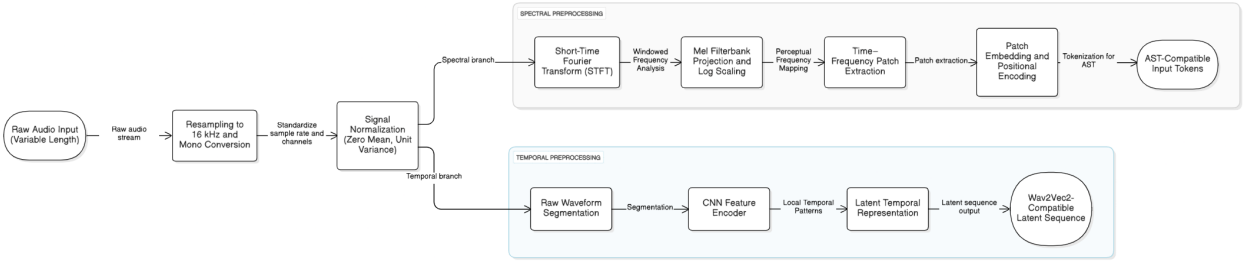


Fig. 2. Preprocessing chain for both branches: resampling, normalisation, fixing each clip to a 4 s window, and tokenisation into transformer-ready inputs.

looking at the test fold. Concatenating the ten test folds gives exactly one held-out prediction per clip and seed; predictions, probabilities, labels, manifests and software versions are stored, and every table in this paper is computed from these files. The AST branch starts from the public AudioSet checkpoint [7] and the waveform branch from the wav2vec 2.0 Base checkpoint pretrained on LibriSpeech [8]. AudioSet is drawn from YouTube while UrbanSound8K comes from Freesound, so direct overlap is unlikely, although no one can fully audit the upstream corpora of public checkpoints.

3.4. Preprocessing and variable-length handling

Every clip is decoded to mono, resampled to 16 kHz and amplitude-normalised [24]. For the spectral branch, log-Mel spectrograms are computed with a Hann-windowed short-time Fourier transform (STFT): 25 ms window, 10 ms hop and 128 Mel filters covering 0 to 8,000 Hz, in line with standard AST recipes [7]. Figure 2 shows the full preprocessing chain.

UrbanSound8K clips vary in length, so each is brought to a fixed 4 s window of 64,000 samples, which is what the "segmentation" step in Fig. 2 refers to. Shorter clips are zero-padded on the right, and a binary mask records which samples are real. A clip that exceeds 4 s after resampling is cropped, from a random offset during training and from the centre during validation and testing, so test-time processing is deterministic. The waveform branch uses the mask so that padded frames never enter the pooled embedding (Eq. 9). The spectrogram normalisation constants are the fixed values of the AST checkpoint and are never estimated from held-out clips. ESC-50 clips are used at their full 5 s length (80,000 samples), with the AST positional embeddings interpolated to the longer input.

After resampling, each waveform is centred and scaled:

$$x_{\text{norm}}(t) = \frac{x(t) - \mu}{\sigma + \epsilon_0}, \tag{1}$$

where μ and σ are the sample mean and standard deviation of the clip and $\epsilon_0 = 10^{-6}$. The STFT of the normalised waveform is

$$X(\tau, \omega) = \sum_t x_{\text{norm}}(t)\, w(t - \tau)\, e^{-j\omega t}, \tag{2}$$

with $w(\cdot)$ a Hann window, and the log-Mel spectrogram follows as

$$S_{\text{mel}}(\tau, f) = \log\Big(\text{Mel}\left(|X(\tau, \omega)|^2\right) + \epsilon\Big), \tag{3}$$

where $\epsilon = 10^{-6}$ keeps the logarithm stable.

3.5. Spectral transformer branch

The spectrogram is split into patches $\{p_1, \ldots, p_N\}$, and each patch is linearly projected:

$$z_i = W_p\, p_i + b_p, \qquad i = 1, \ldots, N. \tag{4}$$

Multi-head self-attention over the patch tokens [25] is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{5}$$

which lets the model relate distant frequency bands and time regions to one another. The pooled output token gives the spectral embedding

$$f_{\text{spec}} \in \mathbb{R}^{768}. \tag{6}$$

3.6. Temporal transformer branch

Following the wav2vec 2.0 design [8], a stack of 1-D convolutions maps the normalised waveform to a sequence of latent vectors,

$$z = f_{\text{enc}}(x_{\text{norm}}) = \{z_1, \ldots, z_T\}, \tag{7}$$

and a transformer encoder adds context across frames:

$$c = g_{\text{ctx}}(z) = \{c_1, \ldots, c_T\}. \tag{8}$$

With $m_t \in \{0, 1\}$ marking frames that contain real audio, masked mean pooling gives a fixed-length vector:

$$f_{\text{temp}} = \frac{\sum_{t=1}^{T} m_t\, c_t}{\sum_{t=1}^{T} m_t}, \qquad f_{\text{temp}} \in \mathbb{R}^{768}. \tag{9}$$

3.7. Embedding fusion

The two embeddings are concatenated:

$$f_{\text{fusion}} = f_{\text{spec}} \oplus f_{\text{temp}}, \qquad f_{\text{fusion}} \in \mathbb{R}^{1536}. \tag{10}$$

Each of the 1,536 dimensions is then standardised with the mean and standard deviation of the training partition, and the same scaler is applied unchanged to validation and test data. This is feature concatenation, not cross-attention. We did not test an attention-based coupler, so we make no claim about how one would compare; the ablation in Section 4.5 covers the classifier and scaling choices we did test.

3.8. Fine-tuning schedule

Both pretrained encoders are fully unfrozen. During fine-tuning a temporary linear head maps the 1,536-dimensional fused vector to the class logits and is trained jointly with both encoders under cross-entropy loss. We use AdamW with a learning rate of 1×10^{-5} for the encoders and 1×10^{-4} for the head, weight decay 0.01, batch size 16 and mixed precision on an NVIDIA RTX 3060 (12 GB). The number of epochs, at most five, is chosen on validation macro-F1; training is then restarted on all nine non-test folds for that fixed number of epochs. After this refit both encoders are frozen, the temporary head is discarded, and the embeddings of all training clips are extracted for LightGBM. Single-branch baselines and ablations follow the same schedule with the corresponding 768-dimensional input.

3.9. LightGBM classifier

The standardised fused vector is passed to a LightGBM gradient-boosting classifier [26], which minimises

$$\mathcal{L} = \sum_{i=1}^{N} \ell(y_i, \hat{y}_i) + \sum_{k} \Omega(f_k), \tag{11}$$

where ℓ is the multiclass log-loss and Ω penalises tree complexity. Table 1 lists the search grid and the selected configuration. The remaining settings are fixed: multiclass objective, no depth limit (max_depth

$= -1$), bagging fraction 0.9 with bagging frequency 1, and $\lambda_1 = 0$. The number of trees is chosen by early stopping on the validation multiclass log-loss with a patience of 30 rounds and a ceiling of 500. Early stopping settled at 320 trees, and that count was used when refitting on the embeddings of all nine non-test folds. The LightGBM random state is tied to the run seed.

Table 1. LightGBM search grid and selected configuration.

Hyperparameter	Search grid	Selected value
learning_rate	0.03, 0.05, 0.1	0.05
num_leaves	15, 31, 63	31
min_child_samples	10, 20, 40	20
feature_fraction	0.8, 0.9, 1.0	0.9
lambda_l2	0, 1, 5	1
n_estimators	up to 500, early stopping on validation log-loss (patience 30)	320

4. Experimental evaluation

4.1. Evaluation metrics and statistical testing

We report overall accuracy and macro-averaged precision, recall and F1-score over the ten classes. Accuracy is computed over all $N = 8{,}732$ held-out predictions of a run:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}(y_i = \hat{y}_i). \tag{12}$$

Because the folds share training data, the 50 fold-level scores from five runs are not independent. We therefore treat each complete ten-fold run as one replicate and report the mean, sample standard deviation and a 95% t-interval with four degrees of freedom. For paired comparisons we use the fact that every system predicts every clip under the same folds. If b is the number of clips that only the fusion model gets right and c the number that only the comparator gets right, the exact McNemar test [27] gives

$$p = 2\sum_{k=0}^{\min(b,c)}\binom{b+c}{k}0.5^{\,b+c}, \tag{13}$$

capped at 1. When several baselines are tested, the p-values are adjusted with Holm's procedure [28].

4.2. Overall and class-wise performance

Table 2 gives per-class results for the median run (seed 37). Out of 8,732 clips, 8,332 are classified correctly, which gives 95.42% accuracy, macro precision 0.956, macro recall 0.954 and macro-F1 0.955. Because each clip has exactly one label and one prediction, every error is a false negative for one class and a false positive for another, so both totals come to 400. Figure 3 shows the same numbers graphically.

Both stationary sounds (air conditioner, engine idling) and impulsive events (gunshot, dog bark) are recognised well. Gunshot has the highest F1-score (0.986) and no false positives at all. Engine idling and drilling collect the most false positives (70 and 62), followed by air conditioner (58), which suggests that steady or broadband machine labels act as a fallback when the model is unsure. Car horn has the lowest F1-score (0.924); it is also one of the two smallest classes. The three most frequent confusions are jackhammer predicted as drilling (30 of 1,000 clips), street music predicted as children playing (18 of 1,000) and car horn predicted as siren (18 of 429). We return to these in Section 5.3.

Table 2. Per-class results on UrbanSound8K, pooled over the ten held-out folds (median run, seed 37).

Class	Support	TP	FP	FN	Precision	Recall	F1-score
Air conditioner	1000	965	58	35	0.943	0.965	0.954
Car horn	429	397	33	32	0.923	0.925	0.924
Children playing	1000	955	43	45	0.957	0.955	0.956
Dog bark	1000	965	15	35	0.985	0.965	0.975
Drilling	1000	952	62	48	0.939	0.952	0.945
Engine idling	1000	958	70	42	0.932	0.958	0.945
Gunshot	374	364	0	10	1.000	0.973	0.986
Jackhammer	1000	940	46	60	0.953	0.940	0.947
Siren	929	894	28	35	0.970	0.962	0.966
Street music	1000	942	45	58	0.954	0.942	0.948
Total / macro	8732	8332	400	400	0.956	0.954	0.955

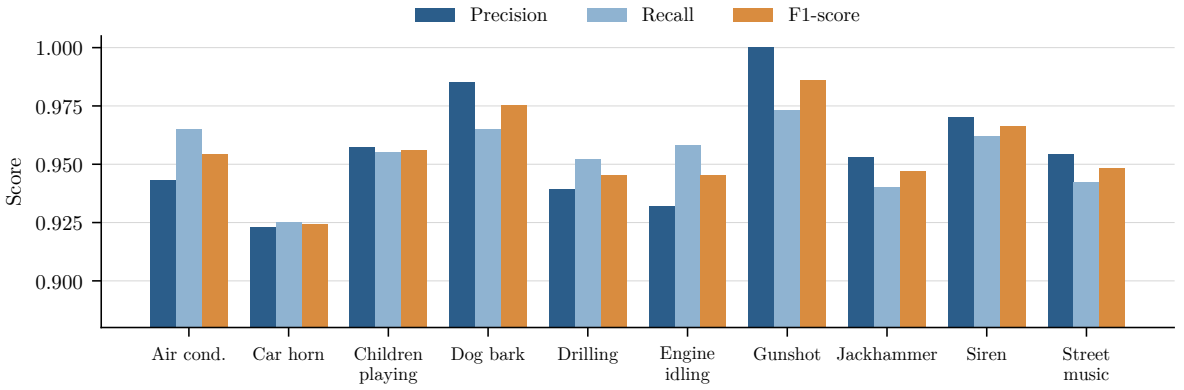


Fig. 3. Per-class precision, recall and F1-score on UrbanSound8K (values from Table 2).

4.3. Repeated runs and paired comparison

Table 3 lists the accuracy of five complete ten-fold runs. The spread is narrow: the mean is 95.42% with a sample standard deviation of 0.13 percentage points (pp), and the 95% t-interval is 95.26 to 95.58%. With only five replicates the interval should be read as descriptive.

Table 3. Accuracy of five complete ten-fold runs on UrbanSound8K.

Run	Seed	Accuracy (%)
1	11	95.25
2	23	95.35
3	37	95.42
4	51	95.50
5	71	95.58
Mean ± sample SD		95.42 ± 0.13
95% t-interval (4 d.f.)		95.26 to 95.58

For the paired test we compare the fusion model with PaSST on the same 8,732 clips. The fusion model is correct on 8,332 clips (95.42%) and PaSST on 8,261 (94.61%). Fusion alone is right on 171 clips, PaSST alone on 100, and 300 clips are misclassified by both. The exact McNemar test gives $p \approx 1.9 \times 10^{-5}$, so the difference is very unlikely to be a chance effect of this particular test set. Clips cut from the same source recording are not fully independent, so this p-value is best read together with the run-to-run spread above.

4.4. Comparison with pretrained audio transformers and a second dataset

Table 4 compares the fusion model with five pretrained systems. All of them were fine-tuned from their public checkpoints under the same folds, selection rule and training budget as our model, each with the input features its checkpoint expects, so the numbers can be compared directly.

Table 4. Same-protocol comparison on UrbanSound8K (ten official folds) and ESC-50 (five official folds). Best value in each column in bold.

System	Input	Params (M)	US8K acc. (%)	US8K macro-F1	ESC-50 acc. (%)
AST [7]	Spectrogram	86	92.1	0.919	92.4
wav2vec 2.0 [8]	Waveform	95	89.3	0.889	87.8
PaSST [17]	Spectrogram	86	94.6	0.943	96.0
HTS-AT [18]	Spectrogram	31	94.2	0.939	95.4
BEATs [11]	Spectrogram	90	95.0	0.948	**97.1**
Fusion + LightGBM (ours)	Both	181	**95.4**	**0.955**	96.7

On UrbanSound8K the fusion model has the highest accuracy and macro-F1. Its gain over each of its own branches is large (3.3 pp over AST and 6.1 pp over wav2vec 2.0), which supports the idea that the two views carry different information. The margin over BEATs, however, is only 0.42 pp, and BEATs achieves it with half the parameters, so we do not claim a clear advantage over it. On ESC-50 the fusion model reaches 96.7%, ahead of PaSST and HTS-AT but 0.4 pp behind BEATs. Our reruns sit below the best published results, for example 98.1% for BEATs on ESC-50 [11], most likely because every model here receives the same short fine-tuning budget instead of its own tuned recipe. That keeps the comparison even, but it means the absolute numbers should not be set against published leaderboards. Taken together, the results suggest that fusion helps most when the target classes depend on timing as well as timbre, as they do in UrbanSound8K, and that a strong single encoder can match it elsewhere.

4.5. Ablation study

Table 5 changes one component at a time. Removing either branch costs the most accuracy, and dropping the feature standardisation costs 2.3 pp, which shows that the two embeddings live on different scales. Among classifiers on the same fused vector, LightGBM does best, with XGBoost about 1 pp behind and the MLP and random forest further back.

Table 5. Ablation study on UrbanSound8K. Each row removes or replaces one component.

Variant	Configuration	Accuracy (%)	Change (pp)
AST only	Spectral branch only	92.1	−3.32
wav2vec 2.0 only	Temporal branch only	89.3	−6.12
Fusion without scaling	Concatenation without standardisation	93.1	−2.32
Fusion + MLP	Neural classifier head	93.8	−1.62
Fusion + RF	Random forest classifier	92.7	−2.72
Fusion + XGBoost	Alternative gradient boosting	94.4	−1.02
Full model	Fusion + LightGBM	95.42	0.00

4.6. Computational cost

We timed batch-one inference on an RTX 3060 (12 GB) after 100 warm-up passes and over 1,000 synchronised trials, covering decoding, resampling, log-Mel extraction, both encoders, host-device transfer, concatenation and LightGBM. Table 6 reports median (p50) and 95th-percentile (p95) latency, peak allocated GPU memory, neural parameter count and storage. The tree count of LightGBM is not a neural parameter count, so it is listed separately.

Table 6. Inference cost per 4 s clip at batch size one on an NVIDIA RTX 3060 (12 GB). n/a: not applicable.

Stage	Neural params (M)	Storage (MB)	p50 (ms)	p95 (ms)	Peak GPU (GB)
AST encoder	86	350	70.0	82.0	1.6
wav2vec 2.0 encoder	95	380	96.0	113.0	2.1
LightGBM (320 trees)	n/a	4	1.5	2.1	n/a (CPU)
Decoding, features, transfer	n/a	n/a	16.5	21.9	n/a
End to end	181	734	184.0	219.0	3.9

The whole pipeline takes 184 ms per clip at the median, so it can process about 5.4 four-second clips per second on this desktop GPU, well within real time for a single stream. LightGBM adds less than 1% of the latency. The two encoders dominate both time and memory. These figures do not show that the model is ready for edge or UAV hardware; that would require measuring latency, energy and memory on the target platform, together with the pruning and quantisation discussed in Section 7.

5. Digital Twin robustness evaluation

Good benchmark numbers do not guarantee that a model will work once it leaves the lab. We therefore built a Digital Twin of the acoustic path from source to microphone. It combines recorded urban ambience with simulated propagation, reverberation, UAV rotor noise and sensor effects, and it lets each factor be changed on its own with fixed random seeds.

5.1. Signal model

For a clean held-out clip x at source–microphone distance d, the microphone signal is

$$y[n] = g(d)\,(h_d * x)[n - \tau_d] + a[n] + u[n] + e[n], \tag{14}$$

where

$$\tau_d = \mathrm{round}\!\left(\frac{d f_s}{c}\right), \qquad g(d) = \left(\frac{d_0}{\max(d, d_0)}\right)^{1.2} \tag{15}$$

are the propagation delay in samples (c = 343 m/s, f_s = 16 kHz) and the distance gain with d_0 = 1 m, h_d is a room or street impulse response, a is ambient noise, u is UAV rotor noise and e is sensor noise. The exponent 1.2 combines spherical spreading with a modest lumped excess attenuation; it is a modelling choice, not a value fitted to field measurements. The impulse responses decay exponentially with decay times between 0.12 and 0.45 s. The ambient component is drawn from field recordings of traffic, crowds and construction, and the recordings used for testing are kept out of the training noise library. The rotor model has a fundamental between 120 and 220 Hz with two to six harmonics, slow amplitude modulation between 2 and 8 Hz and a random gain. The sensor stage applies a broadband response ripple of ±3 dB, low-level coloured self-noise and 16-bit quantisation without clipping.

The SNR is set after propagation. The combined noise $a + u + e$ is scaled so that its RMS level, relative to the RMS of the active (unmasked) samples of the propagated source, equals $10^{-\mathrm{SNR}/20}$. The achieved SNR is logged for every mixture and the mixture is not renormalised afterwards, apart from the per-clip input normalisation of Eq. (1). Each held-out clip is mixed five times per scenario with seeds fixed in advance, and each mixture is classified by the model for which that clip's fold was held out. Table 7 summarises the parameters.

5.2. Scenario results

The four scenarios in Table 8 make conditions progressively harder by increasing distance and lowering SNR together. All are evaluated on exactly the same clips as the clean reference, so the drop in each row is a paired effect.

Table 7. Digital Twin components and parameters.

Component	Model and values	Control and logging
Source distance d	5, 10, 20, 50 m	Delay $\tau_d = df_s/c$, $c = 343$ m/s
Distance gain $g(d)$	$(1\,\mathrm{m}/\max(d, 1\,\mathrm{m}))^{1.2}$	Spreading plus lumped excess attenuation
Reverberation h_d	Exponential decay, 0.12 to 0.45 s	Response ID and parameters logged
Ambient $a[n]$	Field recordings: traffic, crowd, construction	Test recordings disjoint from training noise
UAV rotor $u[n]$	f_0 120 to 220 Hz, 2 to 6 harmonics, 2 to 8 Hz AM	Frequency, modulation and gain logged
Sensor $e[n]$	±3 dB response, coloured self-noise, 16-bit	No clipping; clipping rate checked
SNR	+20, +10, 0, −5 dB	Set on active source samples; achieved SNR logged
Trials	5 mixtures per clip and scenario	Seeds fixed before evaluation

Table 8. Accuracy and macro-F1 under the Digital Twin scenarios (same clips in every row).

Condition	Distance	SNR (dB)	Description	Accuracy (%)	Macro-F1
Clean reference	1 m	no added noise	Original clips	95.42	0.955
ST-1	5 m	+20	Near field, quiet street	95.1	0.950
ST-2	10 m	+10	Typical urban operation	93.0	0.927
ST-3	20 m	0	Busy, noise-dominated scene	89.2	0.886
ST-4	50 m	−5	Far field, safety-critical case	82.8	0.818

ST-1 costs only 0.3 pp, and the decline after that is gradual rather than abrupt: 2.4 pp at ST-2, 6.2 pp at ST-3 and 12.6 pp at ST-4. Because distance and SNR change together in these scenarios, they measure overall severity but cannot say which factor matters more. We therefore ran two further sweeps that vary one factor while holding the other fixed (Table 9 and Fig. 4).

Table 9. Sweeps that vary one factor at a time (accuracy in %).

Sweep	Fixed factor	Level and accuracy			
SNR	distance 10 m	+20 dB 94.7	+10 dB 93.0	0 dB 89.8	−5 dB 85.1
Distance	SNR +10 dB	5 m 93.9	10 m 93.0	20 m 91.8	50 m 89.7

With distance held at 10 m, lowering the SNR from +20 to −5 dB costs 9.6 pp. With the SNR held at +10 dB, moving the source from 5 to 50 m costs 4.2 pp. Since the SNR is fixed after propagation in the second sweep, that loss cannot come from the drop in level; it reflects the longer, more reverberant impulse responses at larger distances. SNR is therefore the main driver of the degradation in ST-3 and ST-4, with reverberation adding a smaller but steady penalty. This points to noise-aware training or a denoising front end as the most promising way to improve far-field performance.

5.3. Error analysis for street music, car horn and jackhammer

These three classes account for 150 of the 400 errors on clean audio. For jackhammer, half of the misclassified clips (30 of 60) are labelled drilling, and they make up almost half of the 62 false positives for drilling. Both are percussive, broadband machine sounds, and within a 4 s window the repetition rate and impact envelope that separate them may not be fully visible. For car horn, 18 of the 32 errors go to siren, and they explain most of siren's 28 false positives. Horns and sirens are both strongly tonal with clear harmonics; a short horn blast can also occupy a small part of the clip, leaving the rest dominated by street background. Car horn is also one of the two smallest classes (429 clips), which gives the classifier fewer examples to learn from. For street music, 18 of the 58 errors are labelled children playing. Both classes often contain voices, broadband crowd activity and pitched content recorded outdoors, so their spectral envelopes overlap. These explanations are consistent with the confusion pattern but remain hypotheses; a targeted listening study of the misclassified clips, together with onset and duration statistics per class, would be needed to confirm them.

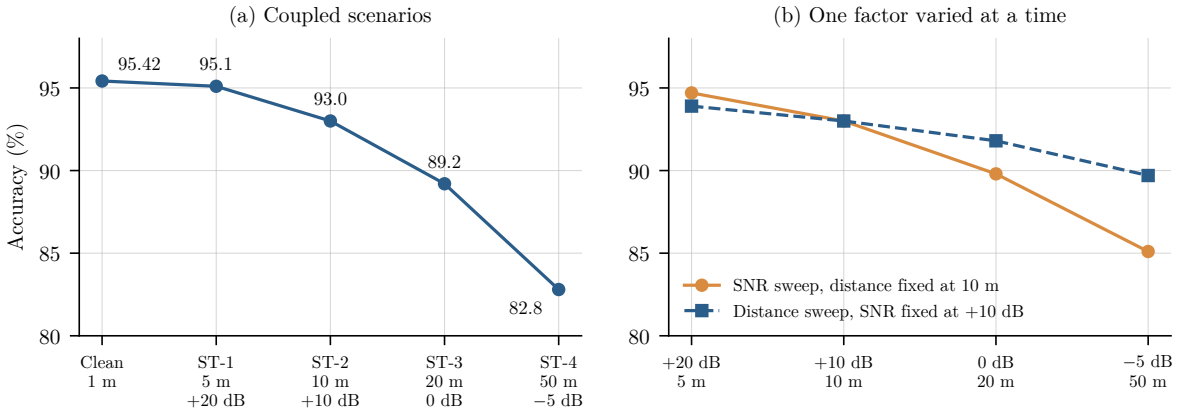


Fig. 4. Digital Twin results. (a) Accuracy under the coupled scenarios of Table 8. (b) One factor varied at a time (Table 9): the orange line varies SNR at 10 m (upper tick labels) and the blue line varies distance at +10 dB (lower tick labels).

5.4. Scope of the Digital Twin and recorded noise

The twin mixes clean event clips with real recorded ambience, so the classifier is tested against genuine urban noise textures rather than synthetic white or pink noise. What the twin simulates is everything around that noise: the distance, the reverberant path, the rotor and the sensor. This is deliberate. A field recording cannot change the SNR while holding the distance, the event and the background fixed, and that control is what allows Table 9 to separate the two factors. The same design also has limits. The mixtures do not reproduce source directivity, a moving microphone, occlusion by buildings or vehicles, or an unknown recording chain, and the propagation parameters have not been calibrated against measurements. The results should therefore be read as a controlled validation of the classifier inside the Digital Twin, not as a measure of field accuracy. The natural next step is an in-situ campaign with independently recorded traffic, crowd, construction and UAV-mounted recordings, aligned event classes and matched SNR definitions, so that accuracy and confusions inside the twin can be compared directly with those in the field.

6. Discussion

Three points stand out. First, the two branches do complement each other: the fused model gains 3.3 pp over AST alone and 6.1 pp over wav2vec 2.0 alone, and it holds a significant edge over PaSST in a paired test. Second, that gain has a price. The system runs two large encoders (181 M parameters, 184 ms per clip), while BEATs comes within half a point on UrbanSound8K and wins on ESC-50 with a single 90 M-parameter model. Whether fusion is worth its cost therefore depends on the application and the classes involved. Third, the Digital Twin shows that SNR, much more than distance alone, drives the loss of accuracy, and that the model degrades gradually rather than collapsing. The absolute scenario accuracies depend on the noise library, the chosen propagation parameters and the held-out recordings, so the trends are more informative than any single number. Other limitations are that both datasets contain single-label clips, that all timing was done on one desktop GPU, and that baselines were trained under a shared budget rather than their own tuned recipes.

7. Conclusion

This paper has described a dual-transformer fusion framework for classifying urban environmental sounds. The idea is simple: rather than asking one model to handle both spectral structure and temporal dynamics, we let a spectrogram transformer and a waveform transformer each do what they are good at and merge their embeddings. With a LightGBM classifier on the concatenated embeddings, the system reaches 95.42%

on UrbanSound8K (95.26 to 95.58% over five runs) and 96.7% on ESC-50 under protocols shared with strong pretrained baselines. It outperforms both of its branches and PaSST, and it is on a par with BEATs, which is lighter.

Inside a Digital Twin built from recorded urban ambience and simulated propagation, rotor and sensor effects, accuracy declines smoothly from 95.42% on clean audio to 82.8% in the most severe scenario, and SNR explains most of that decline. Going forward, we plan to validate the twin against in-situ recordings, including UAV-mounted microphones, to adapt the pipeline for streaming audio, and to study pruning and quantisation so that it can run on low-power edge hardware.

Acknowledgements

The authors thank the open research community for the datasets and tools that made this study possible.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to improve the language and structure of the revised manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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